

Abdullah Munir

Toronto, Canada | 236-457-3130 | abdullahmunir2004@gmail.com | LinkedIn | GitHub | Portfolio

EDUCATION

University of British Columbia

Sep 2022 – June 2026

BSc (Honours) in Computer Science (Co-op) - GPA: 4.20/4.33 (Dean's List)

Kelowna, Canada

- Received UBC's International Major Entrance Scholarship and Faculty of Science International Student Award
- Relevant Courses:** Object-Oriented Programming, Data Structures & Algorithms, Data Analytics, Project Management, Machine Architecture, Artificial Intelligence, Software Engineering, Machine Learning, Parallel Programming (CUDA)

Experience

CGI

June 2026 – Present

Software Engineer (GenAI)

Toronto, Canada

- Built AI migration agents in **Python** converting **Java Swing** into **Angular** and **Spring Boot**, cutting migration time by **60%**.
- Architected **Cloud Run** migration jobs using **GCS FUSE** to read legacy files and write migrated output to Cloud Storage.
- Integrated an open-source LSP-based indexing MCP, led team training on its use, and reduced token consumption by **37%**.
- Built an AI QA workflow that combined legacy code with historical Jira, Bitbucket, and Confluence context to generate Playwright **TypeScript** tests for migrated UI screens.

Software Engineering Intern

Jan 2025 – May 2026

- Deployed open-source LLMs on Vertex AI for internal developer tools, saving **\$4,000** monthly while logging failures.
- Built agentic infrastructure for PostgreSQL and Grafana setup tasks, reducing manual environment configuration.
- Built a prompt library using NestJS to classify AI prompts from Bitbucket/Confluence MCP scans for **45+ teams**.
- Developed a Gemini-powered **RAG** system for Angular UI generation in CGI's Design System, using semantic chunking and hybrid **FAISS/BM25** retrieval to increase developer throughput by approximately **25%**.

University of British Columbia

Sep 2025 – Apr 2026

Machine Learning Engineer

Kelowna, Canada

- Fine-tuned a pretrained **CRAFT** text detector with a **VGG16** backbone for Cree syllabic text detection in **PyTorch**.
- Designed evaluation metrics for Cree syllabic text detection, achieving **88%** accuracy on a limited annotated dataset.
- Ran seeded experiments with validation checkpoints to establish a reproducible Cree text-detection baseline.
- Built a preprocessing pipeline that converted GIMP annotations of Cree documents into character- and word-level CSV labels, providing ground truth for CRAFT training and evaluation.

University of British Columbia, Culture & Technology

May 2024 – Apr 2026

Software Engineer

Kelowna, Canada

- Developed a Unity-based 3D educational game for **24,000+** students using **C#**, Malbers AI, and reusable assets.
- Increased engagement by **30%** using the Observer Pattern to manage state changes for time-based quests.
- Implemented **C#** game-state management to carry player-driven world changes across scenes.

Projects

AllocAid (TA Allocation System) | React, TypeScript, Deno, PostgreSQL, Docker

- Implemented JWT middleware across **100+** routes, enforcing role permissions and blocking cross-account profile edits.
- Built a Deno service running concurrent PostgreSQL checks for availability, schedules, and duplicate assignments.
- Containerized React, Deno, and PostgreSQL with **Docker** Compose and automated CI test coverage in GitHub Actions.

TSKFLO (Task Management System) | React, Redux Toolkit, Node.js, MongoDB, Socket.IO

- Used **RTK Query** cache updates to append Socket.IO messages to active conversations without reloading the page.
- Secured **Socket.IO** connections with JWT handshake middleware and organized live events into conversation rooms.
- Built MongoDB aggregation queries for task status, priority, and top-contributor metrics in admin dashboards.

Technical Skills

Programming Languages: Python, Java, JavaScript, TypeScript, C#, SQL

Frameworks & Libraries: React, Angular, Node.js, NestJS, Spring Boot, Express.js, Unity, PyTorch

Tools & Technologies: Git, Docker, GCP, PostgreSQL, MySQL, MongoDB, REST APIs, Jest, Tableau

Concepts: Full-stack Development, Software Engineering, Cloud Services, Machine Learning, Fine-tuning, Computer Vision, Automation, Context Engineering, CI/CD